HRISHIN DEBNATH

Data Analyst
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SUMMARY

Data analyst and student researcher. I work in SQL, Python, Power BI, Tableau, R, and Excel. I am the sole author of a research paper on Dhaka's air quality (under review). It uses 75,325 hourly readings, and the code is public on GitHub. I am also the sole author of a second paper that forecasts next-day PM2.5 with XGBoost. During internships I built dashboards covering $950M+ in loans and segmented 1,998 retail customers. I am now a research intern at CreateBud, testing how large language models handle medical questions in Bangla. Looking for a data analyst internship.

SKILLS

Programming & Querying: Python (Pandas, NumPy, SciPy, scikit-learn), R, SQL, DAX, JavaScript
Machine Learning: XGBoost, LightGBM, Random Forest, Time-Series Forecasting, Walk-Forward Validation, Optuna
Databases: PostgreSQL, MySQL
BI & Data Visualisation: Power BI, Tableau, Power Query, Microsoft Excel, Matplotlib, Dashboards
Tools & Workflow: Git, GitHub, Power Automate

RESEARCH

Does Dhaka's Air Get Cleaner When the City Empties? A Nine-Year Natural Experiment Using Eid Holidays and Reference-Grade PM2.5 Data
Sole author. Submitted to the Columbia Junior Science Journal (under review) *Sep 2026*

- Measured how 16 Eid holidays (2016 to 2025) affect Dhaka's air, using 75,325 hourly readings from the U.S. Embassy monitor.
- Found PM2.5 falls 36% on average in the holiday window (95% CI 22% to 47%), using an event study and 2,000 placebo runs.
- Made the results reproducible by putting the code, data, and supplement on GitHub.

Next-Day PM2.5 Forecasting in Dhaka, Bangladesh, from a Single Reference-Grade Monitor: A Provenance-Audited Gradient-Boosting Approach Using ERA5 Reanalysis and Open Satellite Fire Data
Sole author. Submitted to Air Quality, Atmosphere & Health (under review) *Sep 2026*

- Beat the persistence baseline by 5.22 µg/m³ RMSE (30.80 vs 36.02, R^2 0.795) on a 12-month holdout with XGBoost.
- Built the pipeline on 248,953 hourly rows by merging OpenAQ, ERA5, NASA POWER, and NASA FIRMS data.
- Checked the result with walk-forward testing and an independent LightGBM model tuned with Optuna. Code on GitHub.

Bangla Gap Study: Large Language Model Accuracy on Medical Questions in Bangla and English (working title)
Researcher, mentored study at CreateBud Inc. In progress; journal submission planned *Jul 2026 - Present*

- Building a Bangla version of 12,000 MedQA questions so the same models can be scored in both languages.
- Chose the translation model and prompt after a 500-item pilot with five machine translators.

WORK EXPERIENCE

CreateBud Inc. **Remote**
Research Intern *Jul 2026 - Present*

- Running the Bangla Gap Study (see Research) with a CreateBud mentor.
- Found 6+ bugs so far in CreateBud's pre-release co-working platform by testing its project, team, and task functions.

VibeWeave (early-stage startup) **Remote**
Market Analyst *Jun 2026 - Present*

- Built textile market studies for five Indian regions: fabric prices, raw materials and listed companies.
- Compiled a verified database of 26 clothing manufacturers.

CORViS **Dhaka, Bangladesh**
Deputy Director, R&D *Dec 2024 - Present*

- Organised 5+ research seminars by approaching professors and industry professionals to speak.
- Built partnerships with 10+ academic and industry experts through direct outreach.

Abelling **Dhaka, Bangladesh**
Data Annotator *Sep 2024 - Apr 2026*

- Annotated 5,000+ datasets for machine learning projects, following detailed labelling guidelines.
- Earned performance bonuses for the work.

Elevvo Pathways **Cairo, Egypt (Remote)**
Data Analyst Intern *Aug 2025 - Sep 2025*

- Found Python and SQL to be the top hiring skills (80%+ prevalence), and Dhaka the main hiring hub, after scraping and analysing 1,500+ job postings.
- Tracked 15% month-over-month growth in peak periods by segmenting the top 20 customers in the Chinook database with SQL.
- Identified 306 Champions who brought in 23.81% of revenue by running RFM segmentation on 1,998 retail customers.

Springer Capital **Chicago, USA (Remote)**
Data Automation Intern *Mar 2025 - Jun 2025*

- Optimised 10+ Power Automate flows to reduce process downtime.
- Made the automated pipelines more accurate by fixing JSON schema and Outlook mailbox configuration issues.
- Improved how interns were evaluated, using data collected from 15+ interns.

TechnoColabs Softwares | LOR & PCC **Indore, India (Remote)**
Data Analyst Intern *Sep 2024 - Nov 2024*

- Reported a 14.93% default rate, 49.66% current status, and 33.42% repayment success across $950M+ in loans with a Power BI dashboard on a company-provided dataset.
- Showed a 16.74% reduction in delinquency was achievable by designing a borrower risk-analysis dashboard.
- Found that term adjustments could cut default risk by 4.4% after analysing repayment trends by credit grade.

PROJECTS

Desk Watch: on-device study-focus app (open source, beta) | **GitHub** *Electron, TypeScript, MediaPipe*

- Windows app that uses the webcam, fully on the device, to ring an alarm when the user picks up a phone, leaves the desk, looks away or falls asleep; also has a site and app blocker, a planner, a Pomodoro focus mode and a Bangla interface.
- Released as open source (MIT) in October 2026, with a Windows installer and 267 unit tests. Co-written with Claude Code: I wrote about 60% of the code, set the requirements, made the design decisions, recorded the training clips and tested each stage.

NASA Space Apps Challenge 2026: Baseline (Team Obsidian Prime) | **GitHub** *Python, JavaScript, HTML, Git*

- Developer on a team building health-monitoring software for astronauts on long-duration missions. Responsible for the codebase and Git repository.
- Built the prescreening prototype, a self-check console that compares each health indicator with the astronaut's own baseline and shows NASA evidence for every flag.

Web Projects *JavaScript, Vercel*

- Built and deployed three live sites: OrigoAid (a crowdfunding platform for Bangladesh), OpAlarm (an opportunity alert app), and Craftly ONE (an NFC card product site).

SaaS Cohort Retention & Scenario Model | GitHub *Excel, Power Query, Power Pivot (DAX)*

- Showed that early churn is the main revenue lever: customer retention fell to 70% by month 12 while net revenue retention held at 95%. Built the macro-free Excel model from a 17,585-row ledger of 2,000 customers over 24 months.
- Estimated 12-month net revenue retention at 83% to 110% by modelling unit economics (ARPA $204, $5,500 gross LTV) and running 1,000 Monte Carlo scenarios.

Bellabeat Wellness Case Study | GitHub *R, SQL*

- Supported personalised wellness recommendations with an analysis of activity data for 450+ users (avg. 6,547 daily steps).
- Suggested app feature improvements after evaluating sleep data for 413 users (avg. 419 minutes).

Olympic Performance Dashboard *Power BI, DAX, Power Query*

- Mapped medal trends across 200+ countries and 120 years of Olympic data in 15+ interactive visualisations.

British Airways Review Analysis | Dashboard *Tableau, Python*

- Split sentiment by service, punctuality, and in-flight experience in a Tableau dashboard built on 10,000+ reviews.
- Made the dashboard clearer by cleaning 2,500+ missing values and standardising text fields.

EDUCATION

St. Joseph Higher Secondary College, Dhaka **2026 - 2028 (expected)**
Higher Secondary Certificate (HSC), Science, English version

WorldQuant University **Dec 2025 - Jun 2026**
Applied Data Science Lab

CERTIFICATIONS

Google Data Analytics Professional Certificate **Sep 2024 - Nov 2024**
Coursera

Google Business Intelligence Professional Certificate **Nov 2024 - Dec 2024**
Coursera

Other Certifications: Data Analytics Essentials (Cisco Networking Academy), Data Analysis with Python (freeCodeCamp), Advanced Power BI: DAX Language and Formulas (LinkedIn), SQL (Advanced) Exam (HackerRank)
Job Simulations (Forage): Tata - Data Visualisation, Accenture - Data Analytics and Visualization, Quantium - Data Analytics